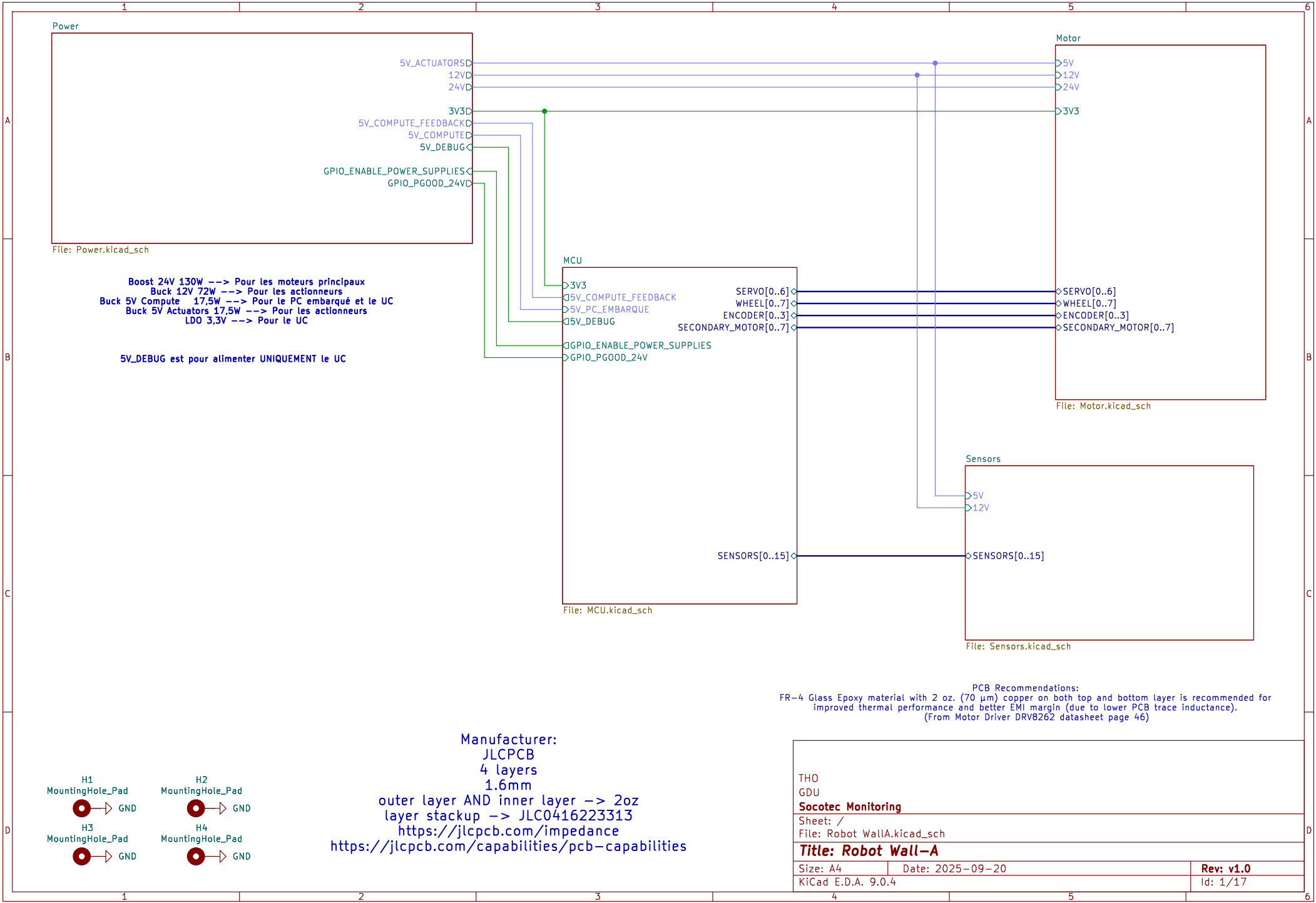




Power

Motor

Sensors

MCU

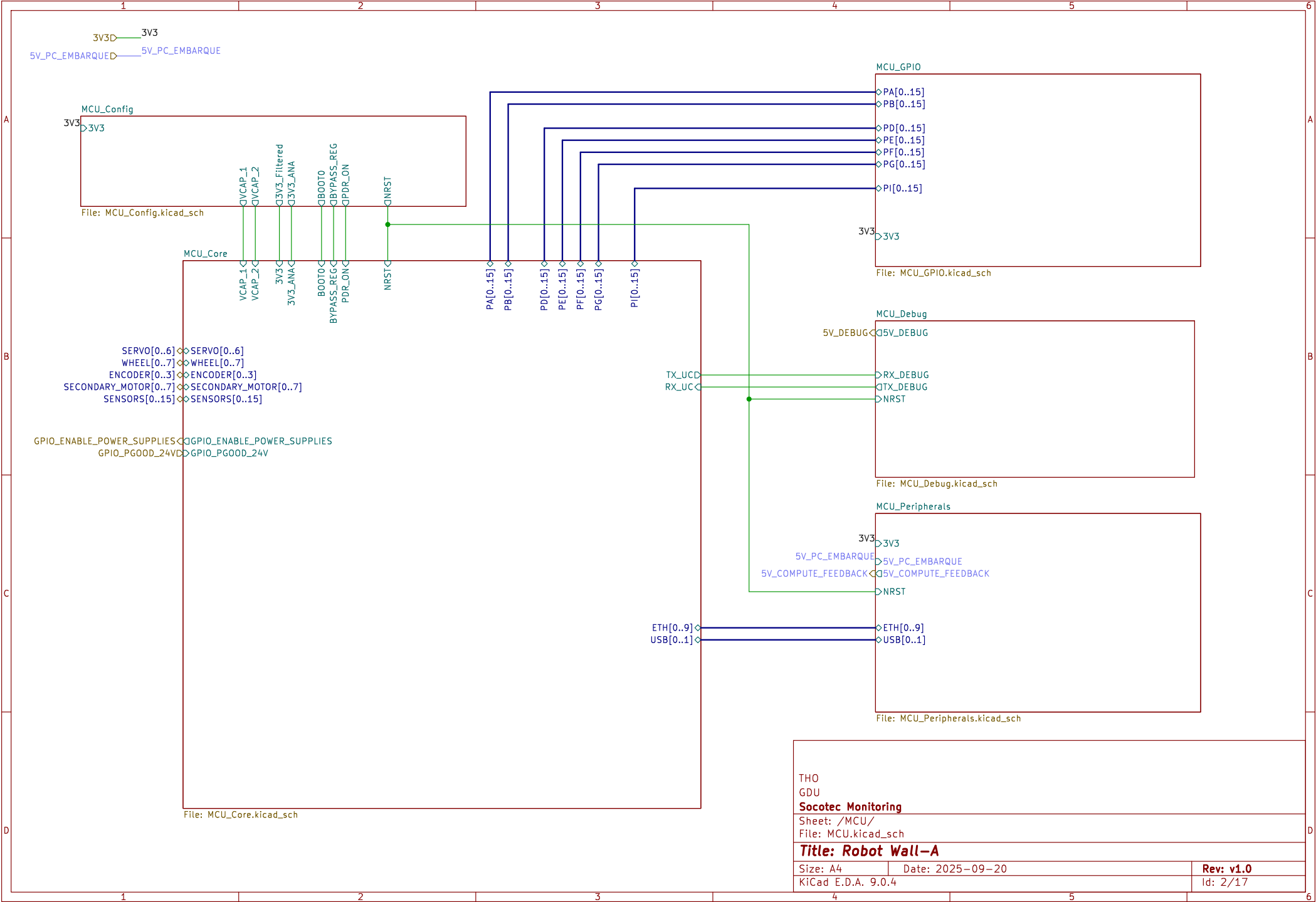
File: Power.kicad_sch

File: Motor.kicad_sch

File: MCU.kicad_sch

File: Sensors.kicad_sch

PCB Recommendations:
FR-4 Glass Epoxy material with 2 oz. (70 μ m) copper on both top and bottom layer is recommended for improved thermal performance and better EMI margin (due to lower PCB trace inductance).
(From Motor Driver DRV8262 datasheet page 46)



TH0
GDU

Socotec Monitoring

Sheet: /MCU/
File: MCU.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20

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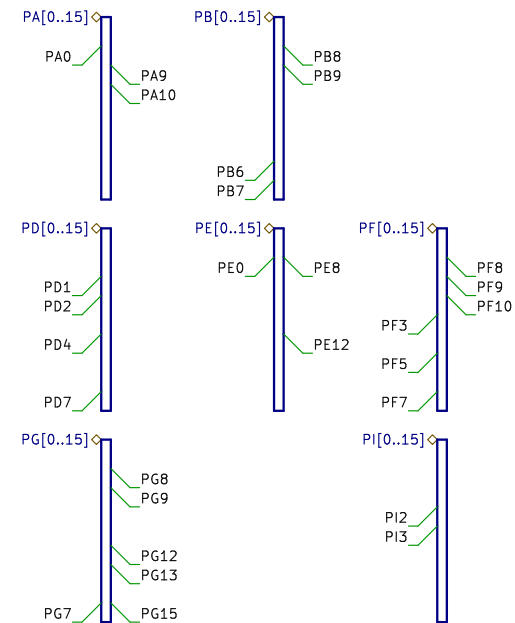
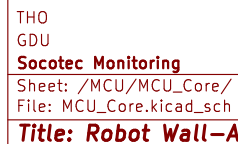
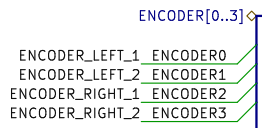
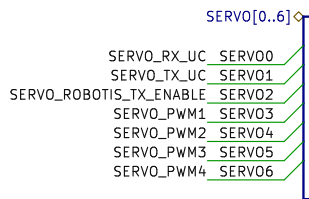
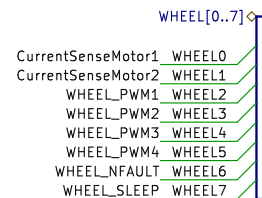
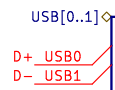


Diagram showing the connection of the Ethernet PHY to the Ethernet MAC. The PHY signals are connected to the MAC signals, which are then connected to the Ethernet controller signals.

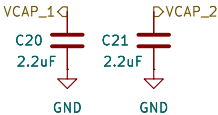
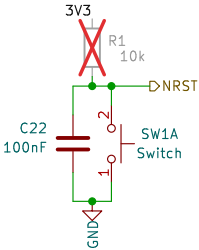
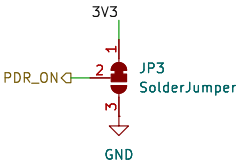
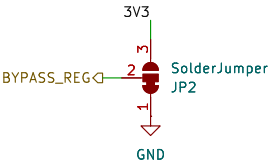
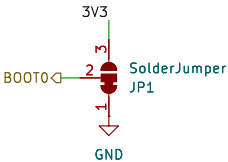
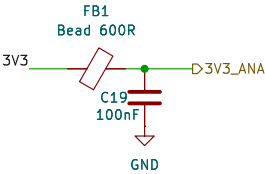
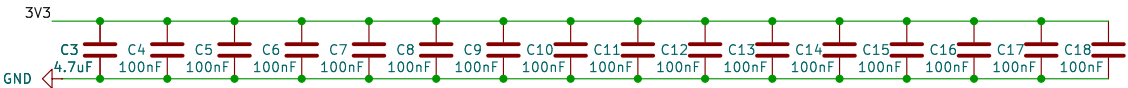
PHY Signal	MAC Signal	Ethernet Controller Signal
RMII_CRX_DV	ETH0	ETH8 RMII_TXD0
RMII_RXD0	ETH1	ETH9 RMII_TXD1
RMII_RXD1	ETH2	
RMII_MDIO	ETH3	
RMII_REF_CLK	ETH4	
RMII_MDC	ETH5	
RMII_TX_EN	ETH6	
PHY_CLK_25MHZ	ETH7	



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Must be as close as possible from μC

3V3 $\rightarrow$ 3V3_Filtered



TH0
GDU

Socotec Monitoring
Sheet: /MCU/MCU_Config/
File: MCU_Config.kicad_sch

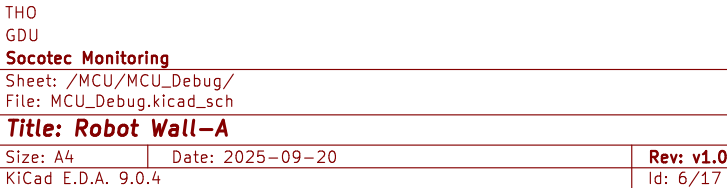
Title: Robot Wall-A

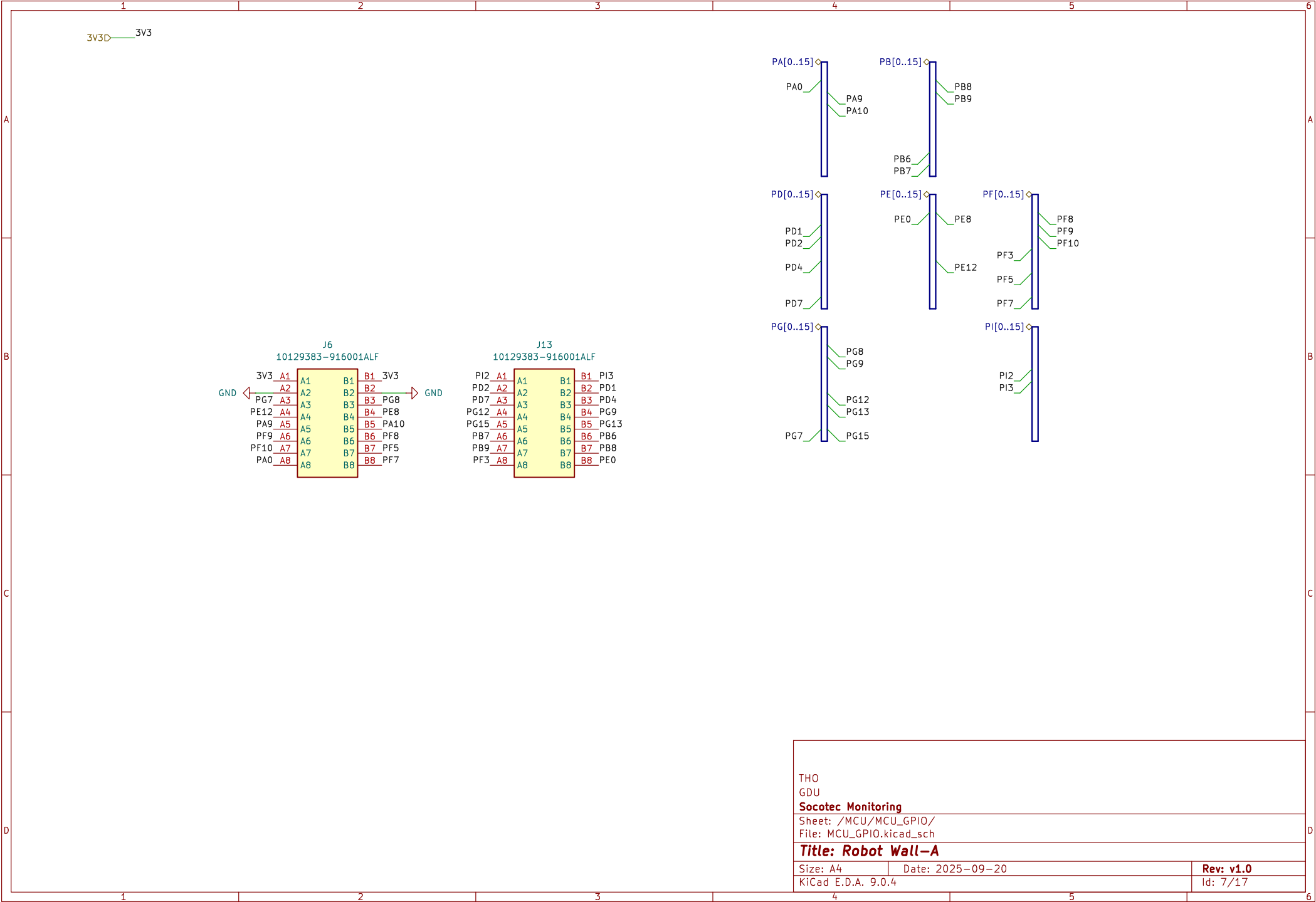
Size: A4
Date: 2025-09-20

KiCad E.D.A. 9.0.4

Rev: v1.0
Id: 4/17

[illegible]





TH0

GDU

Socotec Monitoring

Sheet: /MCU/MCU_GPIO/

File: MCU_GPIO.kicad_sch

Title: Robot Wall-A

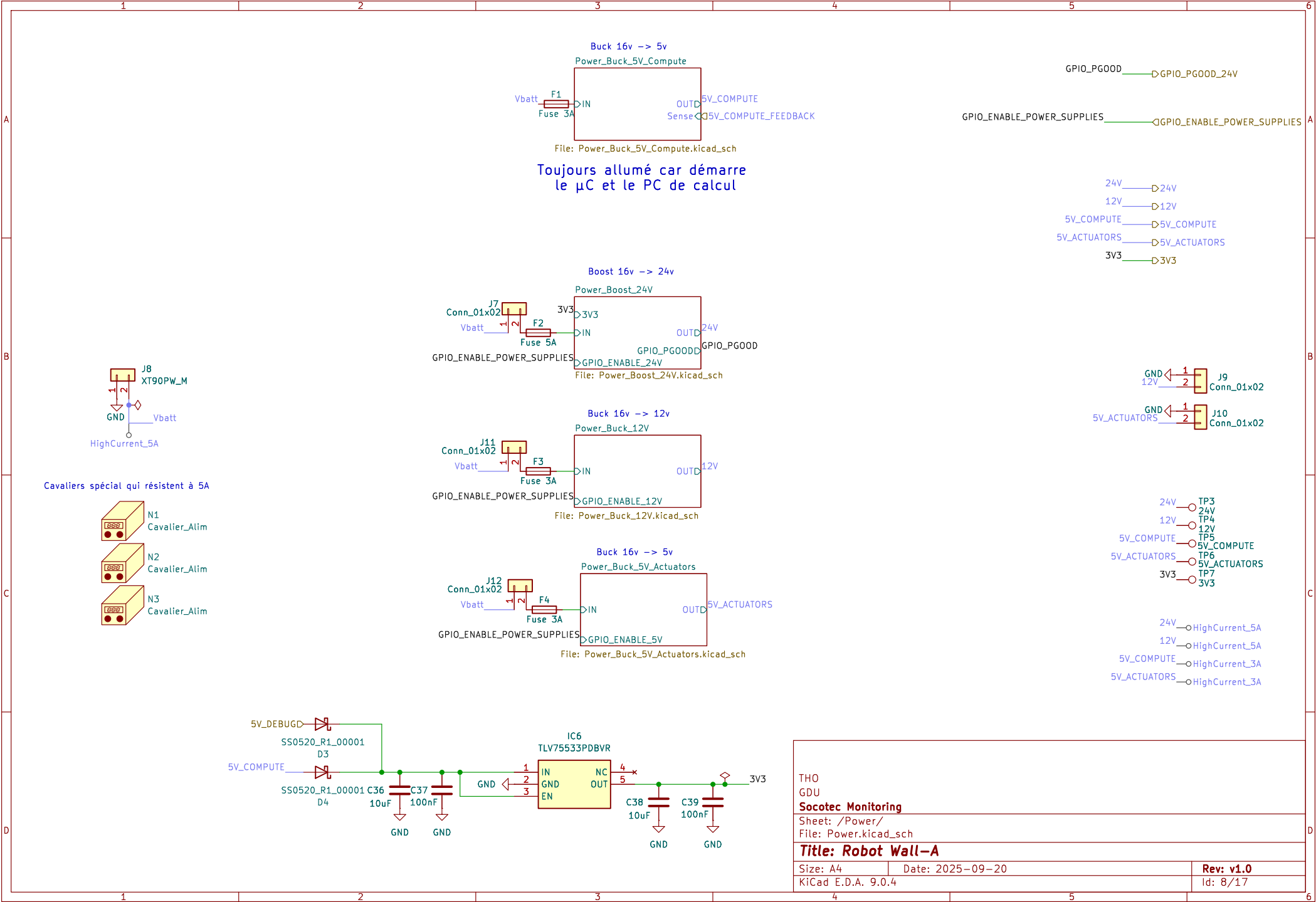
Size: A4

Date: 2025-09-20

Rev: v1.0

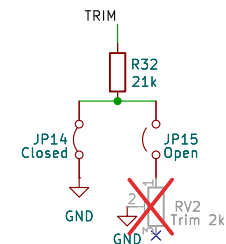
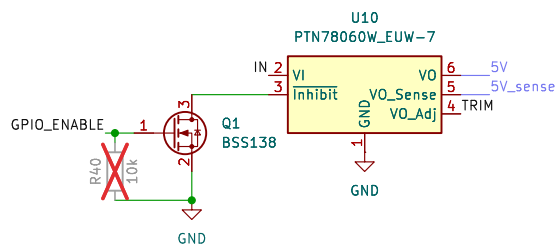
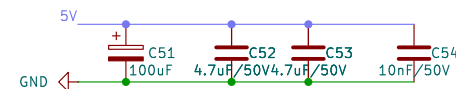
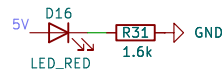
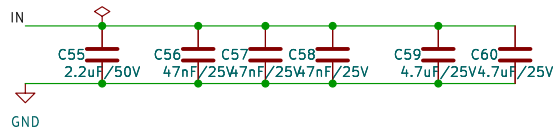
KiCad E.D.A. 9.0.4

Id: 7/17



IND IN

5V OUT
5V_sense Sense



TH0
GDU

Socotec Monitoring

Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

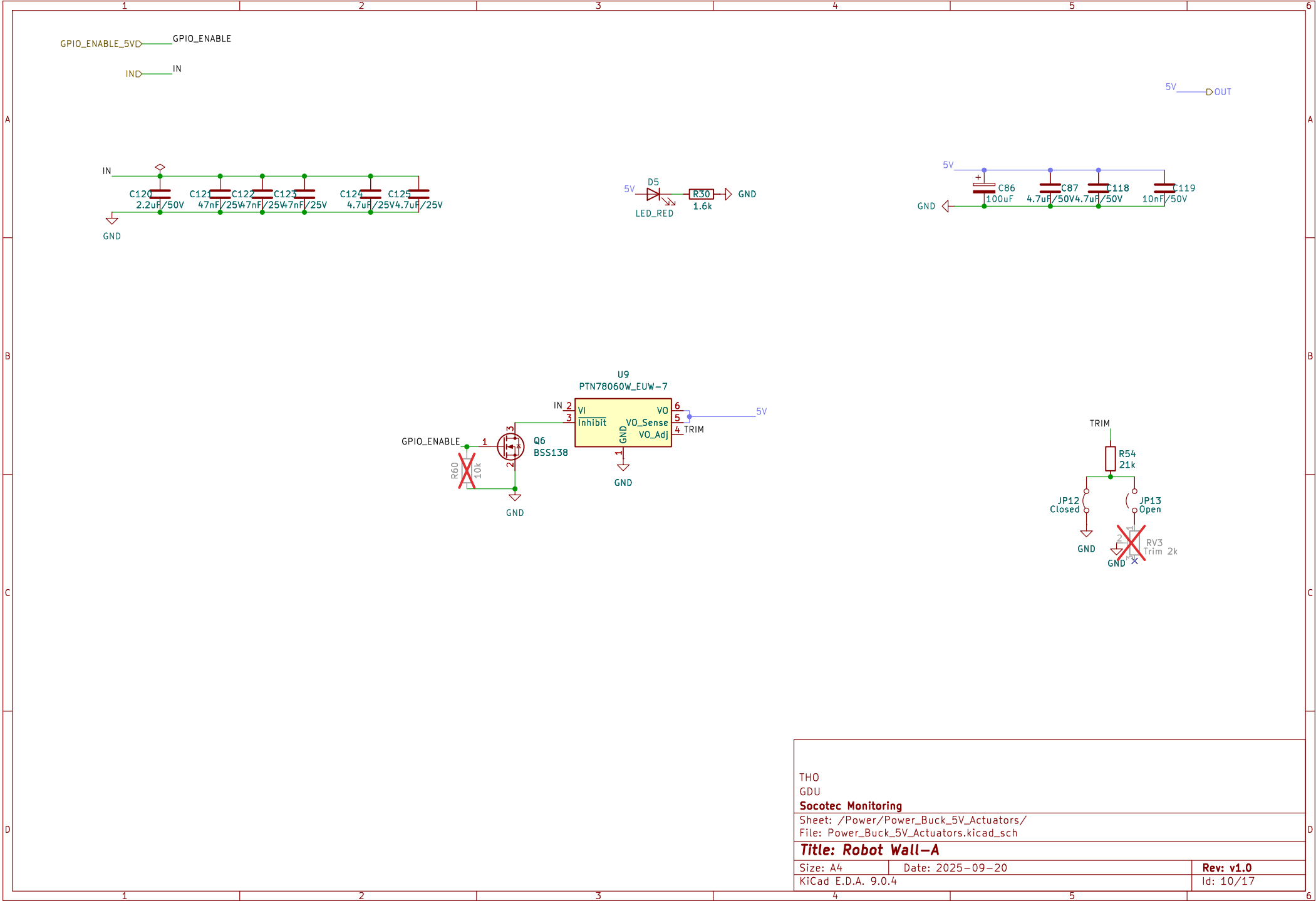
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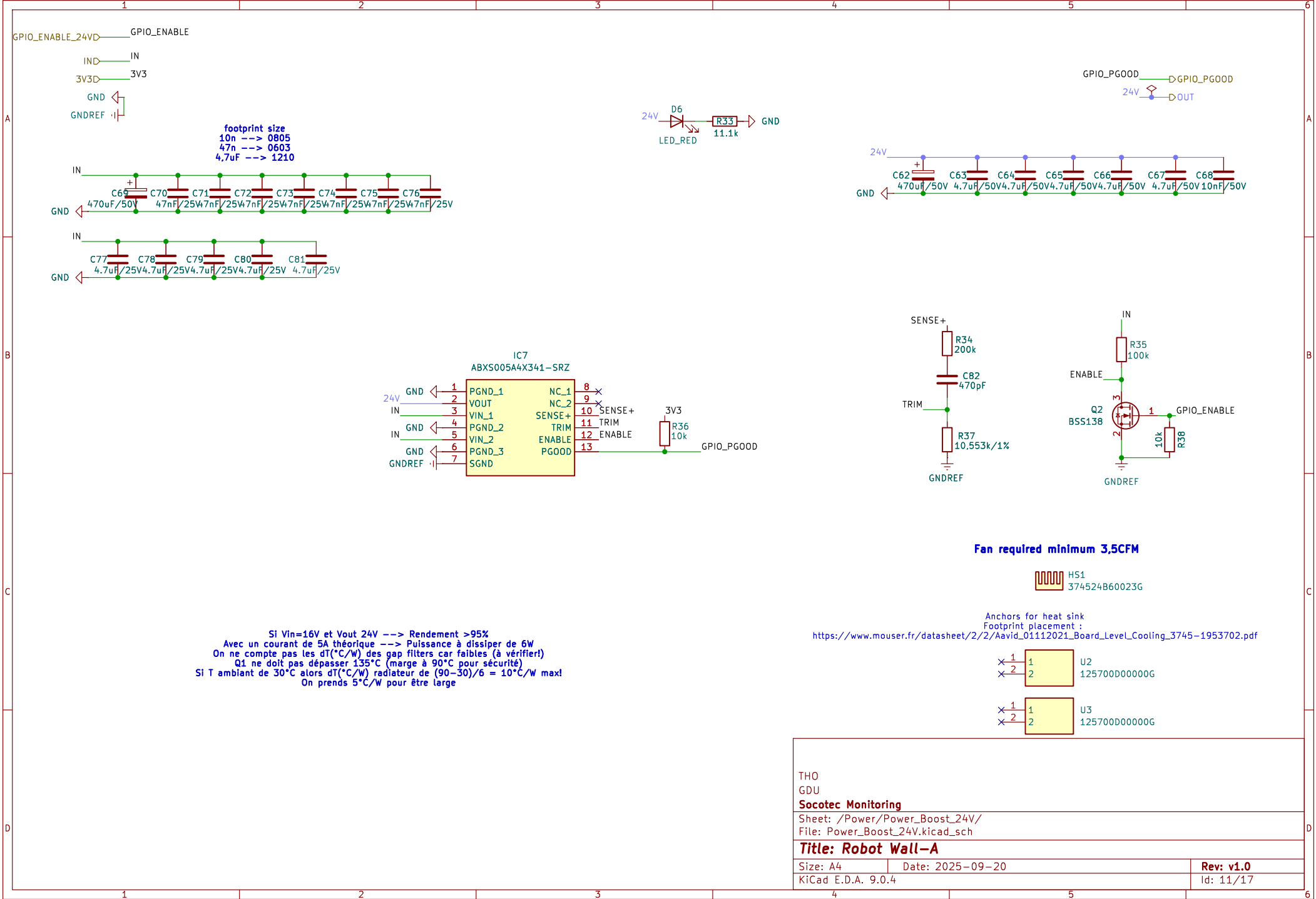
Size: A4 Date: 2025-09-20

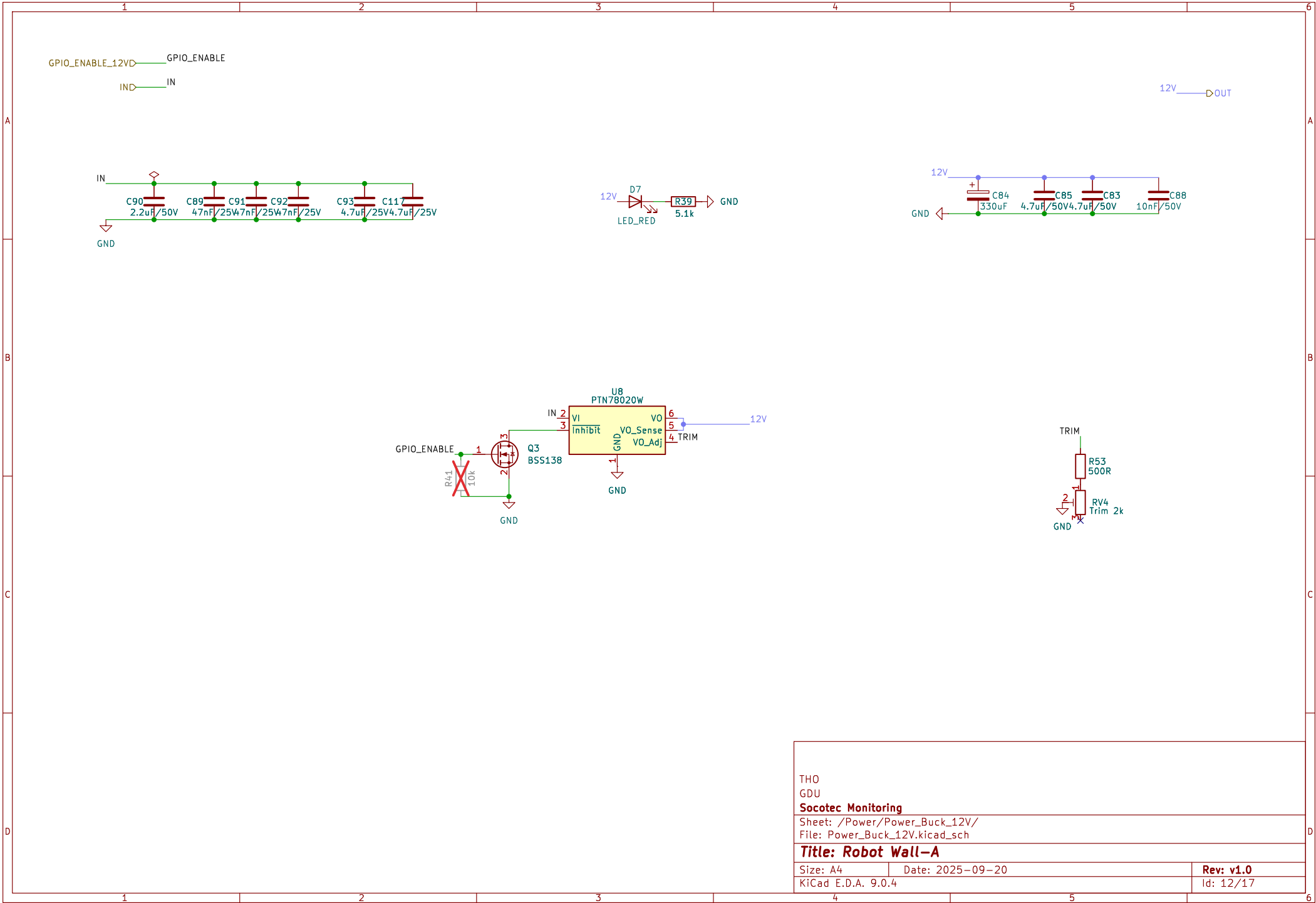
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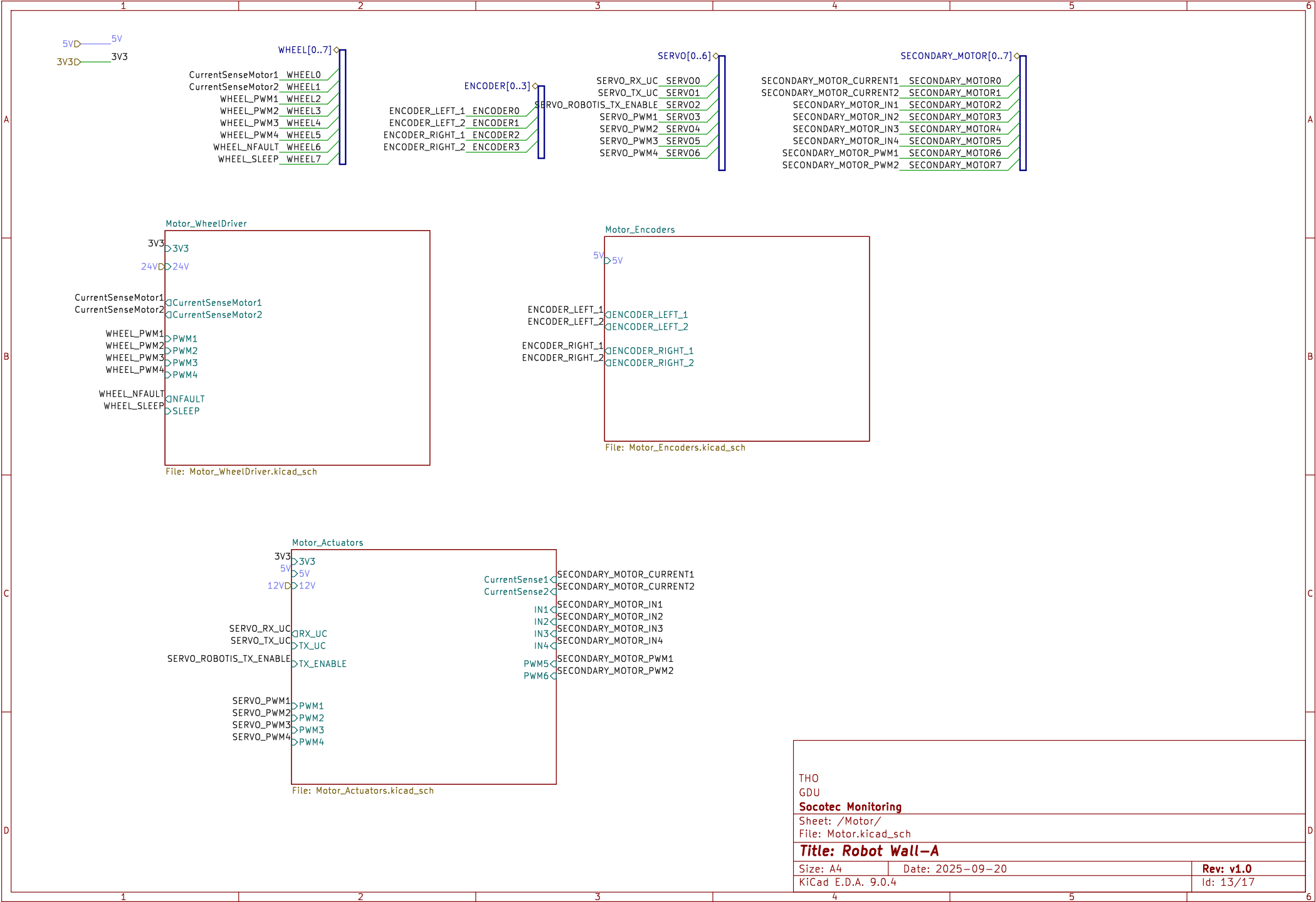
Rev: v1.0

Id: 9/17







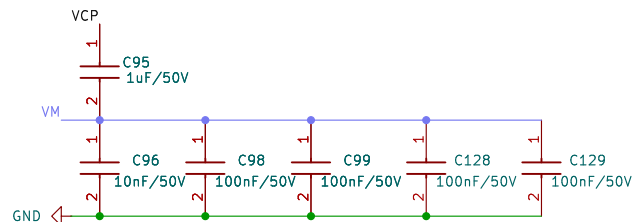
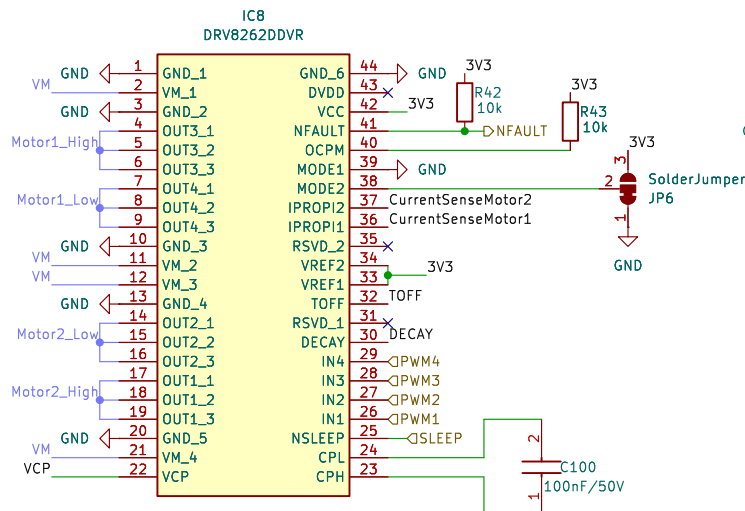


3V3D 3V3
24VD VM



Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

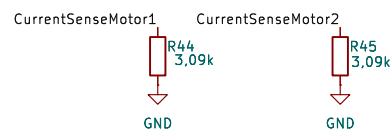
CurrentSenseMotor1 --> CurrentSenseMotor1
CurrentSenseMotor2 --> CurrentSenseMotor2



Motor1_High 1 1 J16
Motor1_Low 2 2 39501-1002

Motor2_Low 1 1 J15
Motor2_High 2 2 39501-1002

Current limit to 5A (example page 21 datasheet)



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

U4
125700D00000G

U5
125700D00000G

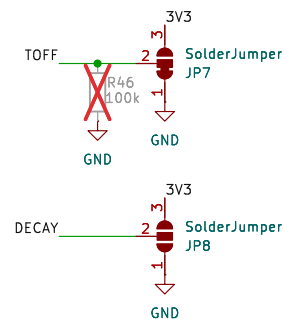
HS2
374524B60023G

Fan required minimum 3,5CFM

2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High --> HighCurrent_5A
Motor1_Low --> HighCurrent_5A
Motor2_High --> HighCurrent_5A
Motor2_Low --> HighCurrent_5A

TH0
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

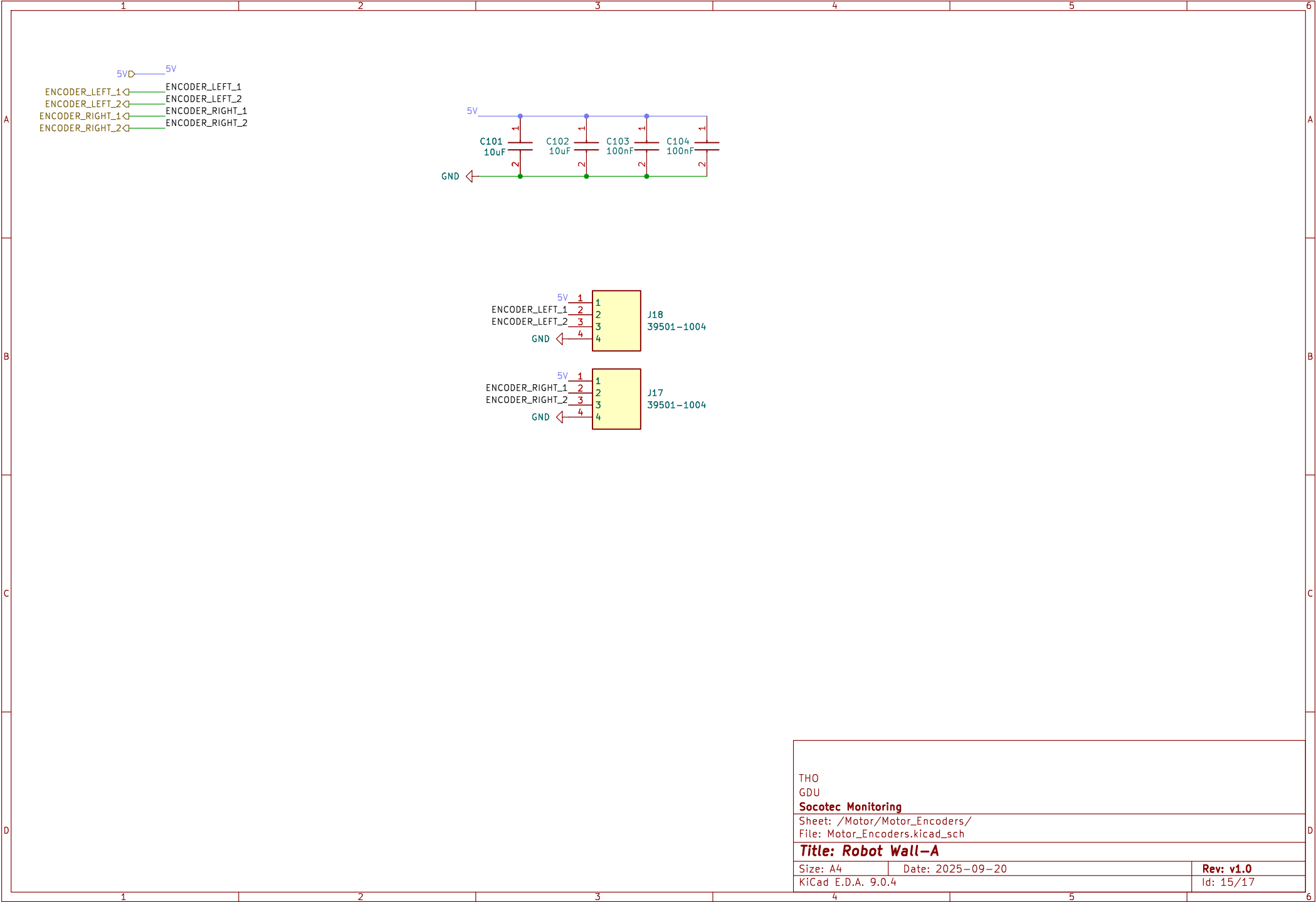
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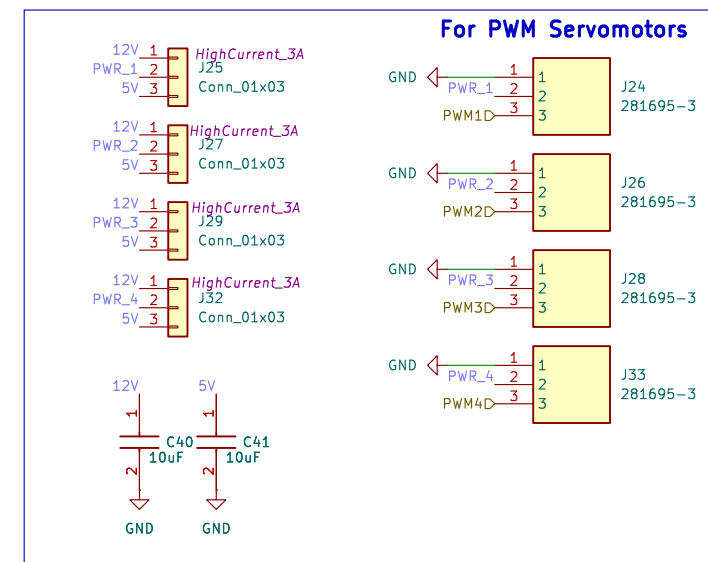
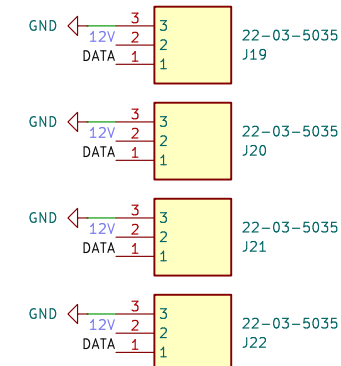
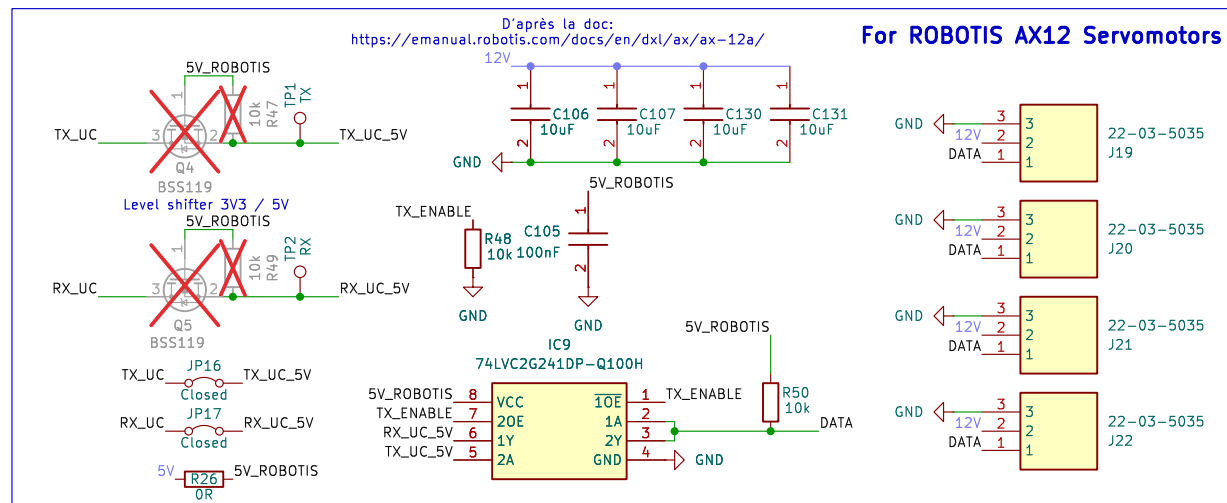
Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

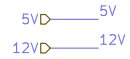
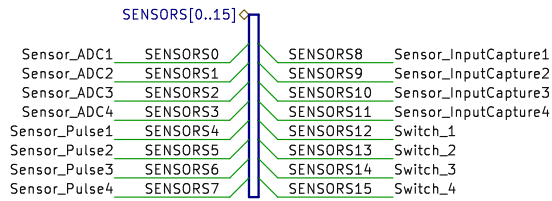
Rev: v1.0

Id: 14/17



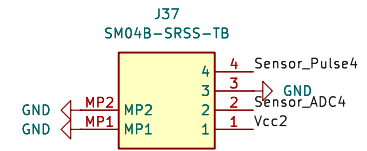
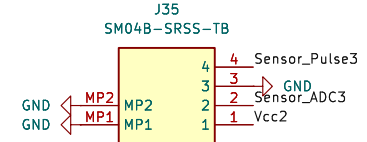
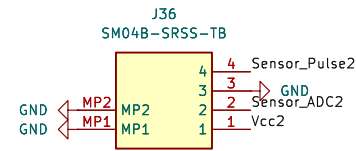
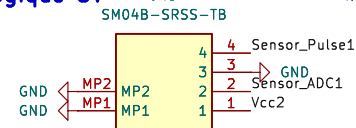
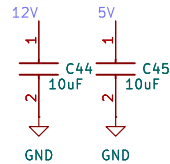
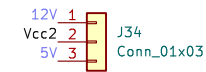


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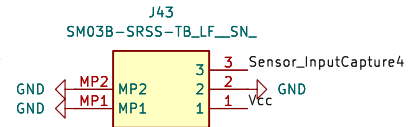
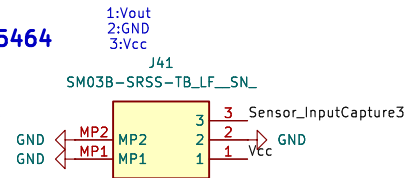
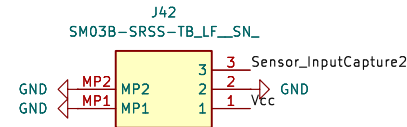
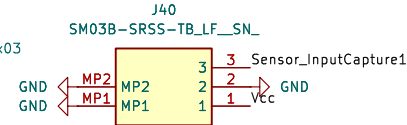
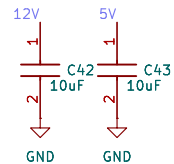
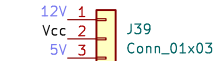
Capteurs de distance

Pour capteur B5W-LB2101-1 analogique 5V

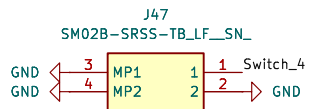
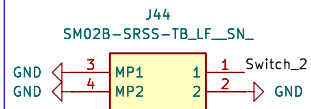
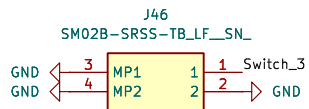
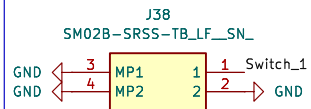


https://omronfs.omron.com/en_US/ecb/products/pdf/en-b5w_lb_series_users_manual.pdf

Pour capteur Pololu <https://www.pololu.com/product/5464>



Interrupteurs standard



TH0
GDU

Socotec Monitoring

Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20

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Rev: v1.0

Id: 17/17